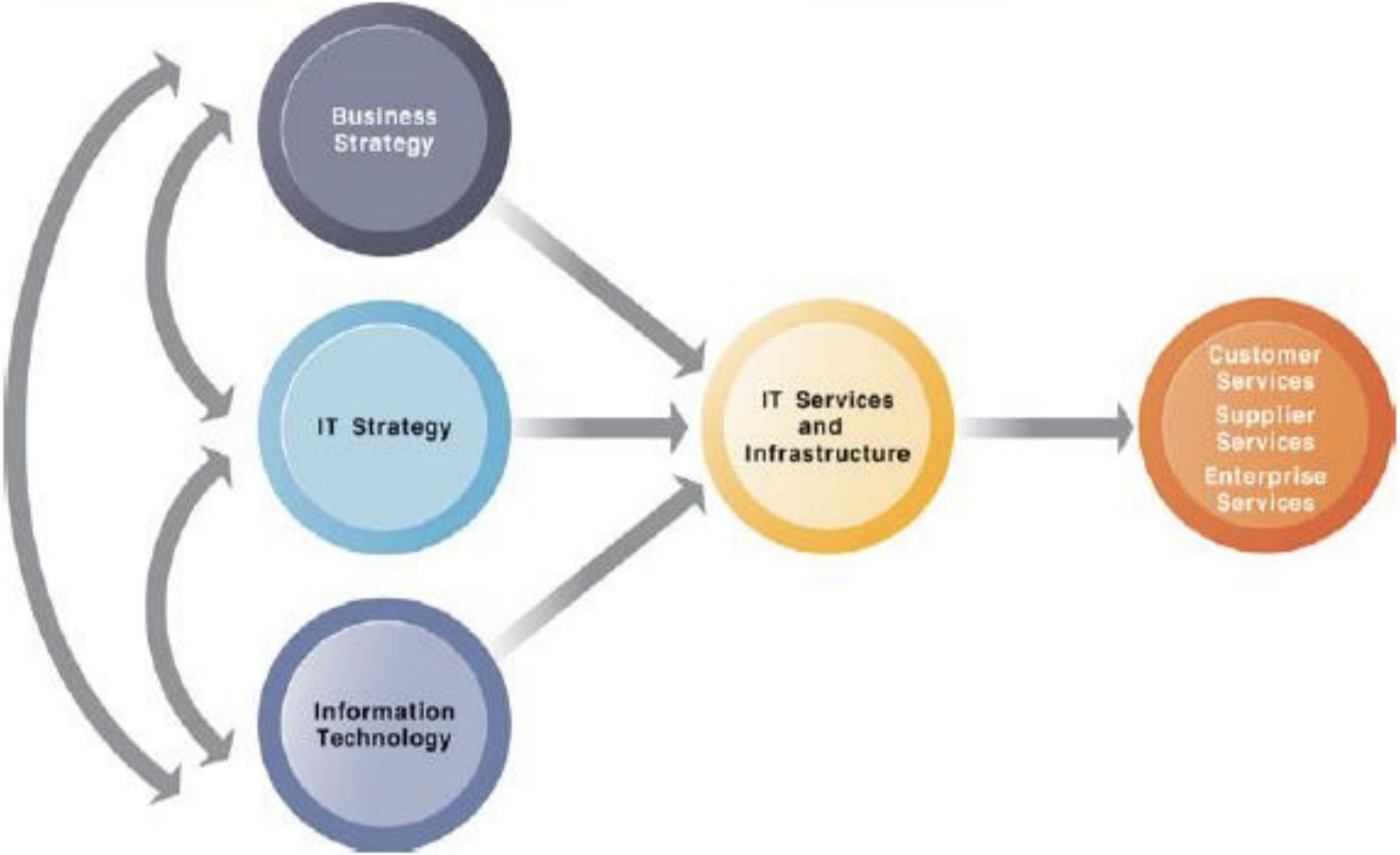
INFORMATION TECHNOLOGY INFRASTRUCTURE

Unit-III

IT INFRASTRUCTURE

- IT infrastructure includes investment in hardware, software, and services—such as consulting, education, and training—that are shared across the entire firm or across entire business units in the firm.

- A firm's IT infrastructure provides the foundation for serving customers, working with vendors, and managing internal firm business processes.



The services a firm is capable of providing to its customers, suppliers, and employees are a direct function of its IT infrastructure. Ideally, this infrastructure should support the firm's business and information systems strategy. New information technologies have a powerful impact on business and IT strategies, as well as the services that can be provided to customers.

DEFINING IT INFRASTRUCTURE:

- IT infrastructure consists of a set of physical devices and software applications that are required to operate the entire enterprise.
- But IT infrastructure is also a set of firmwide services budgeted by management and comprising both human and technical capabilities.
- These services include the following:

1. **Computing platforms** used to provide computing services that connect employees, customers, and suppliers into a coherent digital environment, including large mainframes, midrange computers, desktop and laptop computers, and mobile handheld devices.
2. **Telecommunications services** that provide data, voice, and video connectivity to employees, customers, and suppliers.
3. **Data management services** that store and manage corporate data and provide capabilities for analyzing the data.
4. **Application software services** that provide enterprise-wide capabilities such as enterprise resource planning, customer relationship management, supply chain management, and knowledge management systems that are shared by all business units.

DEFINING IT INFRASTRUCTURE:

5. **Physical facilities management services** that develop and manage the physical installations required for computing, telecommunications, and data management services.
6. **IT management services** that plan and develop the infrastructure, coordinate with the business units for IT services, manage accounting for the IT expenditure, and provide project management services.
7. **IT standards services** that provide the firm and its business units with policies that determine which information technology will be used, when, and how.
8. **IT education services** that provide training in system use to employees and offer managers training in how to plan for and manage IT investments.
9. **IT research and development services** that provide the firm with research on potential future IT projects and investments that could help the firm differentiate itself in the market place.

COMPONENTS OF IT INFRASTRUCTURE

1. Hardware

   Hardware includes servers, datacenters, personal computers, routers, switches, and other equipment.

2. Software

   Software refers to the applications used by the business, such as web servers, content management systems, and the OS—like Linux. The OS is responsible for managing system resources and hardware, and makes the connections between all of your software and the physical resources that do the work.

3. Networking

   Interconnected network components enable network operations, management, and communication between internal and external systems. The network consists of internet connectivity, network enablement, firewalls and security, as well as hardware like routers, switches, and cables.

   e.g., LAN, WAN, VPN

HARDWARE PLATFORM TRENDS

- **THE MOBILE DIGITAL PLATFORM**
- Smartphones such as the iPhone, Android, and BlackBerry smartphones have taken on many functions of PCs, including transmission of data, surfing the Web, transmitting e-mail and instant messages, displaying digital content, and exchanging data with internal corporate systems.
- The new mobile platform also includes small, lightweight netbooks optimized for wireless communication and Internet access, tablet computers such as the iPad, and digital e-book readers such as Amazon's Kindle with Web access capabilities.

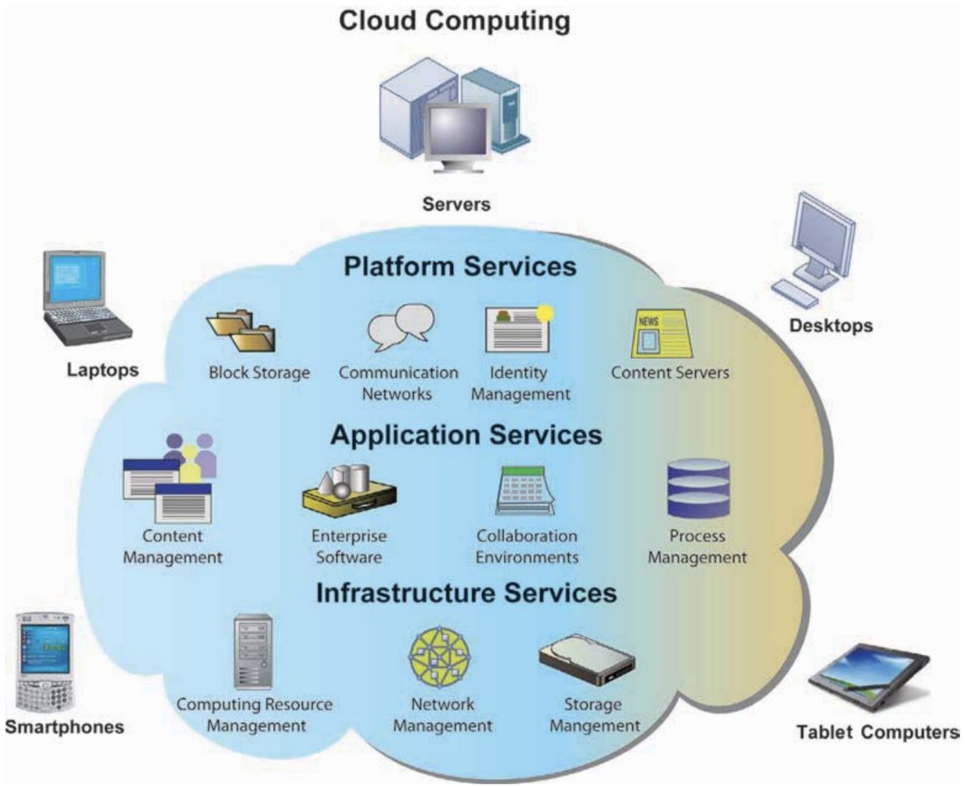
HARDWARE PLATFORM TRENDS

- **CONSUMERIZATION OF IT AND BYOD**
- The popularity, ease of use, and rich array of useful applications for smartphones and tablet computers have created a groundswell of interest in allowing employees to use their personal mobile devices in the workplace, a phenomenon popularly called "bring your own device" (BYOD). BYOD is one aspect of the consumerization of IT, in which new information technology that first emerges in the consumer market spreads into business organizations.
- Consumerization of IT includes not only mobile personal devices but also business uses of software services such as Google and Yahoo search, Gmail, Google Apps, Dropbox and even Facebook and Twitter that originated in the consumer marketplace as well.

HARDWARE PLATFORM TRENDS

- **CLOUD COMPUTING**
- Cloud computing is a model of computing in which computer processing, storage, software, and other services are provided as a pool of virtualized resources over a network, primarily the Internet. These “clouds” of computing resources can be accessed on an as-needed basis from any connected device and location

CLOUD COMPUTING



In cloud computing, hardware and software capabilities are a pool of virtualized resources provided over a network, often the Internet. Businesses and employees have access to applications and IT infrastructure anywhere, at any time, and on any device.

GRID COMPUTING

- **Grid computing** involves connecting geographically remote computers into a single network to create a virtual supercomputer by combining the computational power of all computers on the grid
- Grid computing was impossible until high-speed Internet connections enabled firms to connect remote machines economically and move enormous quantities of data. Grid computing requires software programs to control and allocate resources on the grid.

VIRTUALIZATION

- Virtualization is the process of presenting a set of computing resources (such as computing power or data storage) so that they can all be accessed in ways that are not restricted by physical configuration or geographic location.
- Virtualization enables a single physical resource (such as a server or a storage device) to appear to the user as multiple logical resources.
- For example, a server or mainframe can be configured to run many instances of an operating system so that it acts like many different machines.
- Virtualization also enables multiple physical resources (such as storage devices or servers) to appear as a single logical resource, as would be the case with storage area networks or grid computing.
- Virtualization makes it possible for a company to handle its computer processing and storage using computing resources housed in remote locations. VMware is the leading virtualization software vendor for Windows and Linux servers.

GREEN COMPUTING

- Green computing or green IT, refers to practices and technologies for designing, manufacturing, using, and disposing of computers, servers, and associated devices such as monitors, printers, storage devices, and networking and communications systems to minimize the impact on the environment.
- Reducing computer power consumption has been a very high "green" priority

HIGH-PERFORMANCE AND POWER-SAVING PROCESSORS

- Another way to reduce power requirements and hardware sprawl is to use more efficient and power-saving processors.
- Microprocessors now feature multiple processor cores (which perform the reading and execution of computer instructions) on a single chip.
- A multicore processor is an integrated circuit to which two or more processor cores have been attached for enhanced performance, reduced power consumption, and more efficient simultaneous processing of multiple tasks.
- This technology enables two or more processing engines with reduced power requirements and heat dissipation to perform tasks faster than a resource-hungry chip with a single processing core.
- Today you’ll find PCs with dual-core, quad-core, six-core, and eight core processors.

AUTONOMIC COMPUTING

- With large systems encompassing many thousands of networked devices, computer systems have become so complex today that some experts believe they may not be manageable in the future.
- One approach to this problem is autonomic computing. Autonomic computing is an industry-wide effort to develop systems that can configure themselves, optimize and tune themselves, heal themselves when broken, and protect themselves from outside intruders and self-destruction.
- You can glimpse a few of these capabilities in desktop systems. For instance, virus and firewall protection software are able to detect viruses on PCs, automatically defeat the viruses, and alert operators.
- These programs can be updated automatically as the need arises by connecting to an online virus protection service such as McAfee. IBM and other vendors are starting to build autonomic features into products for large systems.

SOFTWARE PLATFORM TRENDS

- There are three major themes in software platform evolution:
  - Linux and open source software
  - Java, HTML, and HTML5
  - Web services and service-oriented architecture

LINUX AND OPEN SOURCE SOFTWARE

- Open source software is software produced by a community of several hundred thousand programmers around the world.
- According to the leading open source professional association, OpenSource.org, open source software is free and can be modified by users.
- Works derived from the original code must also be free, and the software can be redistributed by the user without additional licensing.
- Open source software is by definition not restricted to any specific operating system or hardware technology, although most open source software is currently based on a Linux or Unix operating system.

LINUX

- Perhaps the most well-known open source software is Linux, an operating system related to Unix.
- Linux was created by the Finnish programmer Linus Torvalds and first posted on the Internet in August 1991.
- Linux applications are embedded in cell phones, smartphones, netbooks, and consumer electronics.
- Linux is available in free versions downloadable from the Internet.
- The rise of open source software, particularly Linux and the applications it supports, has profound implications for corporate software platforms: cost reduction, reliability and resilience, and integration, because Linux works on all the major hardware platforms from mainframes to servers to clients.

SOFTWARE FOR THE WEB: JAVA, HTML, AND HTML5

- Java is an operating system-independent, processor-independent, object-oriented programming language that has become the leading interactive environment for the Web. Java was created by James Gosling and the Green Team at Sun Microsystems in 1992.
- Java developers can create small applet programs that can be embedded in Web pages and downloaded to run on a Web browser. A Web browser is an easy-to-use software tool with a graphical user interface for displaying Web pages and for accessing the Web and other Internet resources.
- Java is being used for more complex e-commerce and e-business applications that require communication with an organization's back-end transaction processing systems.

HTML AND HTML5

- HTML (Hypertext Markup Language) is a page description language for specifying how text, graphics, video, and sound are placed on a Web page and for creating dynamic links to other Web pages and objects.
- Using these links, a user need only point at a highlighted keyword or graphic, click on it, and immediately be transported to another document.
- The next evolution of HTML, called HTML5

Features	Html	Html5
definition	A hypertext markup language (HTML) is the primary language for developing web pages.	HTML5 is a new version of HTML with new functionalities with markup language with Internet technologies.
Multimedia support	Language in **HTML** does not have support for video and audio.	**HTML5** supports both video and audio.
Storage	The HTML browser uses cache memory as temporary storage.	HTML5 has the storage options like:**application cache, SQL database,** and **web storage**.
Browser compatibility	HTML is compatible with almost all browsers because it has been present for a long time, and the browser made modifications to support all the features.	In HTML5, we have many new tags, elements, and some tags that have been **removed/modified**, so only some browsers are fully compatible with **HTML5**.
Graphics support	In HTML, vector graphics are possible with tools Like**Silver light, Adobe Flash, VML,** etc.	In HTML5, vector graphics are supported by default.
Threading	In HTML, the browser interface and JavaScript running in the same thread.	The HTML5 has the JavaScript Web Worker API, which allows the browser interface to run in multiple threads.
Storage	Uses cookies to store data.	Uses local storage instead of cookies
Vector and Graphics	Vector graphics are possible with the help of technologies like **VML, Silverlight, Flash,etc**.	Vector graphics is an integral part of **HTML5, SVG** and **canvas**.
Shapes	It is not possible to create shapes like **circles, rectangles, triangles**.	We can draw shapes like **circles, rectangles, triangles**.
Multimedia support	Audio and video are not the part of HTML4.	Audio and video are essential parts of HTML5,like: **<Audio>, <Video>**.
Vector Graphics	In HTML4, vector graphics are possible with the help of techniques like VML, Silver light and Flash.	Vector graphics are an integral part of **HTML5, SVG**, and **canvas**.
Shapes	It is not possible to draw shapes like circles, rectangles, triangles.	Using html5, you can draw shapes like **circles, rectangles, triangles**.

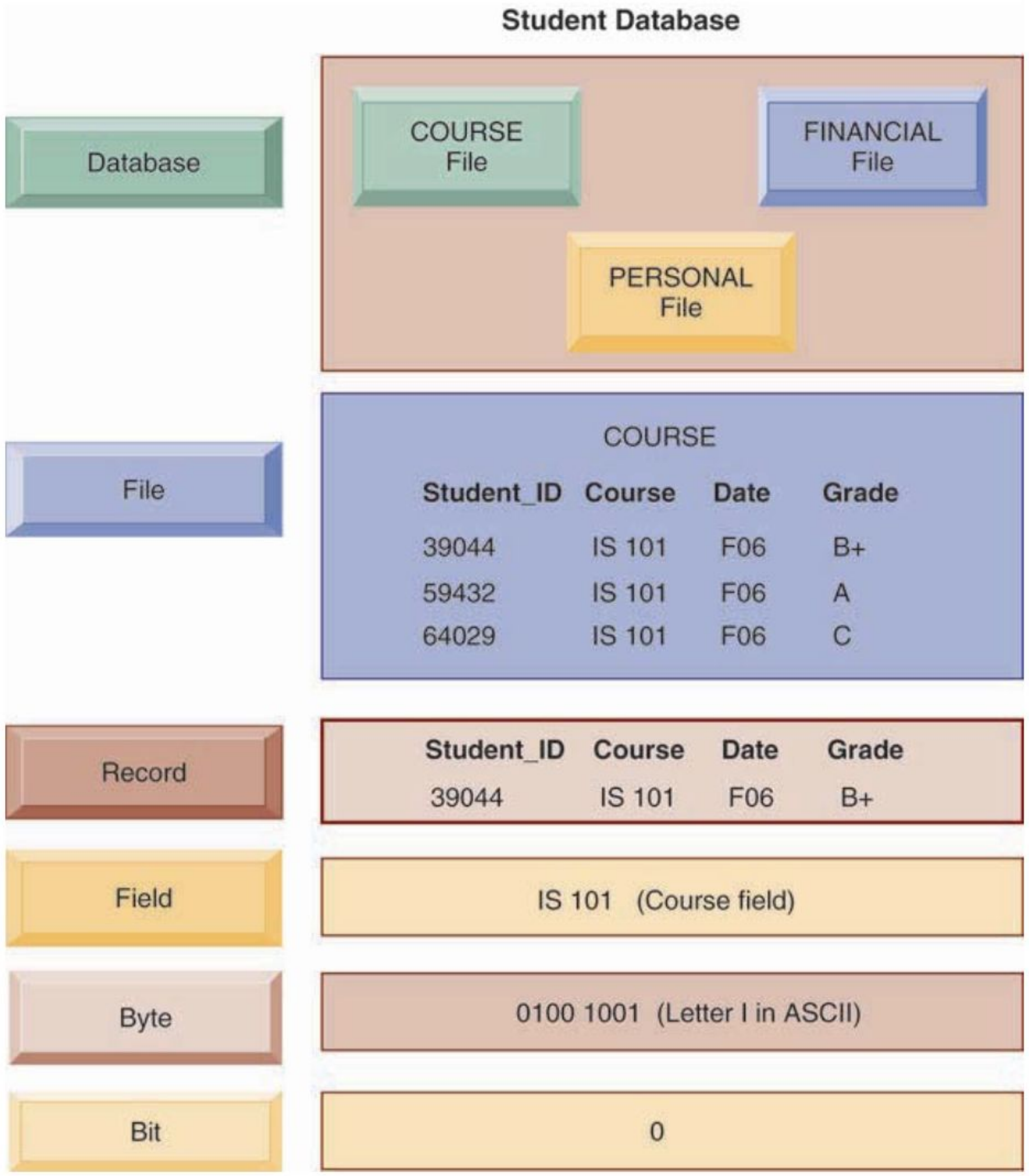
WEB SERVICES AND SERVICE-ORIENTED ARCHITECTURE

- Web services refer to a set of loosely coupled software components that exchange information with each other using universal Web communication standards and languages.
- They can exchange information between two different systems regardless of the operating systems or programming languages on which the systems are based.
- They can be used to build open standard Web-based applications linking systems of two different organizations, and they can also be used to create applications that link disparate systems within a single company.
- Web services are not tied to any one operating system or programming language, and different applications can use them to communicate with each other in a standard way without time-consuming custom coding.
- The foundation technology for Web services is XML, which stands for Extensible Markup Language. This language was developed in 1996 by the World Wide Web Consortium as a more powerful and flexible markup language than hypertext markup language (HTML) for Web pages.
- Whereas HTML is limited to describing how data should be presented in the form of Web pages, XML can perform presentation, communication, and storage of data

SERVICE - ORIENTED ARCHITECTURE

- The collection of Web services that are used to build a firm's software systems constitutes what is known as a service-oriented architecture.
- A service- oriented architecture (SOA) is set of self-contained services that communicate with each other to create a working software application.
- Business tasks are accomplished by executing a series of these services. Software developers reuse these services in other combinations to assemble other applications as needed.
- Virtually all major software vendors provide tools and entire platforms for building and integrating software applications using Web services.
- IBM includes Web service tools in its WebSphere e-business software platform, and Microsoft has incorporated Web services tools in its Microsoft .NET platform.

FIGURE 6.1 THE DATA HIERARCHY

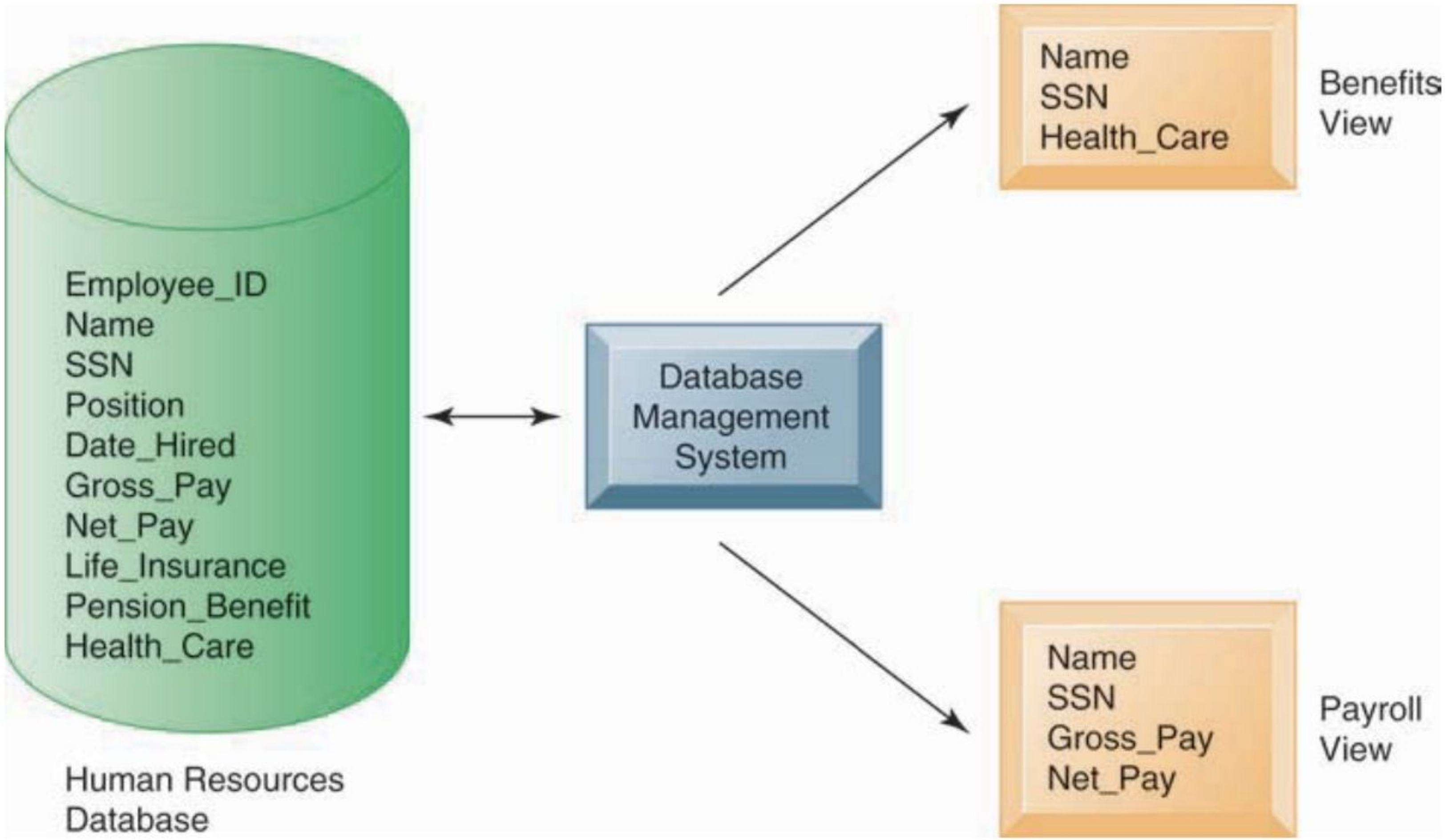


A computer system organizes data in a hierarchy that starts with the bit, which represents either a 0 or a 1. Bits can be grouped to form a byte to represent one character, number, or symbol. Bytes can be grouped to form a field, and related fields can be grouped to form a record. Related records can be collected to form a file, and related files can be organized into a database.

DATABASE MANAGEMENT SYSTEM

- A database management system (DBMS) is software that permits an organization to centralize data, manage them efficiently, and provide access to the stored data by application programs.
- The DBMS acts as an interface between application programs and the physical data files.
- When the application program calls for a data item, such as gross pay, the DBMS finds this item in the database and presents it to the application program.
- Using traditional data files, the programmer would have to specify the size and format of each data element used in the program and then tell the computer where they were located.

FIGURE 6.3 HUMAN RESOURCES DATABASE WITH MULTIPLE VIEWS



A single human resources database provides many different views of data, depending on the information requirements of the user. Illustrated here are two possible views, one of interest to a benefits specialist and one of interest to a member of the company's payroll department.

RELATIONAL DBMS

- DBMS use different database models to keep track of entities, attributes, and relationships.
- The most popular type of DBMS today for PCs as well as for larger computers and mainframes is the relational DBMS. Relational databases represent data as two-dimensional tables (called relations).
- Tables may be referred to as files.
- Each table contains data on an entity and its attributes.

FIGURE 6.4 RELATIONAL DATABASE TABLES

SUPPLIER

Columns (Attributes, Fields)

Supplier_Number	Supplier_Name	Supplier_Street	Supplier_City	Supplier_State	Supplier_Zip
8259	CBM Inc.	74 5th Avenue	Dayton	OH	45220
8261	B. R. Molds	1277 Gandolly Street	Cleveland	OH	49345
8263	Jackson Composites	8233 Micklin Street	Lexington	KY	56723
8444	Bryant Corporation	4315 Mill Drive	Rochester	NY	11344

Rows (Records, Tuples)

Key Field (Primary Key)

PART

Part_Number	Part_Name	Unit_Price	Supplier_Number
137	Door latch	22.00	8259
145	Side mirror	12.00	8444
150	Door molding	6.00	8263
152	Door lock	31.00	8259
155	Compressor	54.00	8261
178	Door handle	10.00	8259

Primary Key

Foreign Key

A relational database organizes data in the form of two-dimensional tables. Illustrated here are tables for the entities SUPPLIER and PART showing how they represent each entity and its attributes. Supplier_Number is a primary key for the SUPPLIER table and a foreign key for the PART table.

- The actual information about a single supplier that resides in a table is called a row. Rows are commonly referred to as records, or in very technical terms, as tuples. Data for the entity PART have their own separate table.
- The field for Supplier_Number in the SUPPLIER table uniquely identifies each record so that the record can be retrieved, updated, or sorted. It is called a key field.
- Each table in a relational database has one field that is designated as its primary key. This key field is the unique identifier for all the information in any row of the table and this primary key cannot be duplicated.
- Supplier_ Number is the primary key for the SUPPLIER table and Part_Number is the primary key for the PART table. Note that Supplier_Number appears in both the SUPPLIER and PART tables. In the SUPPLIER table, Supplier_Number is the primary key. When the field Supplier_Number appears in the PART table, it is called a foreign key and is essentially a lookup field to look up data about the supplier of a specific part.

FIGURE 6.5 THE THREE BASIC OPERATIONS OF A RELATIONAL DBMS

PART

Part_Number	Part_Name	Unit_Price	Supplier_Number
137	Door latch	22.00	8259
145	Side mirror	12.00	8444
150	Door molding	6.00	8263
152	Door lock	31.00	8259
155	Compressor	54.00	8261
178	Door handle	10.00	8259

Select Part_Number = 137 or 150

SUPPLIER

Supplier_Number	Supplier_Name	Supplier_Street	Supplier_City	Supplier_State	Supplier_Zip
8259	CBM Inc.	74 5th Avenue	Dayton	OH	45220
8261	B. R. Molds	1277 Gandolly Street	Cleveland	OH	49345
8263	Jackson Components	8233 Micklin Street	Lexington	KY	56723
8444	Bryant Corporation	4315 Mill Drive	Rochester	NY	11344

Join by Supplier_Number

Part_Number	Part_Name	Supplier_Number	Supplier_Name
137	Door latch	8259	CBM Inc.
150	Door molding	8263	Jackson Components

Project selected columns

The select, join, and project operations enable data from two different tables to be combined and only selected attributes to be displayed.

QUERYING AND REPORTING

- DBMS includes tools for accessing and manipulating information in databases. Most DBMS have a specialized language called a data manipulation language that is used to add, change, delete, and retrieve the data in the database. This language contains commands that permit end users and programming specialists to extract data from the database to satisfy information requests and develop applications. The most prominent data manipulation language today is Structured Query Language, or SQL.
- Microsoft Access and other DBMS include capabilities for report generation so that the data of interest can be displayed in a more structured and polished format than would be possible just by querying. Crystal Reports is a popular report generator for large corporate DBMS, although it can also be used with Access. Access also has capabilities for developing desktop system applications. These include tools for creating data entry screens, reports, and developing the logic for processing transactions.

FIGURE 6.7 EXAMPLE OF AN SQL QUERY

```
SELECT PART.Part_Number, PART.Part_Name, SUPPLIER.Supplier_Number,
SUPPLIER.Supplier_Name
FROM PART, SUPPLIER
WHERE PART.Supplier_Number = SUPPLIER.Supplier_Number AND
Part_Number = 137 OR Part_Number = 150;
```

Illustrated here are the SQL statements for a query to select suppliers for parts 137 or 150. They produce a list with the same results as Figure 6.5.

BUSINESS INTELLIGENCE INFRASTRUCTURE

- Data Warehouses and Data Marts
- Hadoop
- In-Memory Computing

DATA WAREHOUSE

- A data warehouse is a database that stores current and historical data of potential interest to decision makers throughout the company.
- The data originate in many core operational transaction systems, such as systems for sales, customer accounts, and manufacturing, and may include data from Web site transactions.
- The data warehouse extracts current and historical data from multiple operational systems inside the organization.
- These data are combined with data from external sources and transformed by correcting inaccurate and incomplete data and restructuring the data for management reporting and analysis before being loaded into the data warehouse.

DATA MART

- A data mart is a subset of a data warehouse in which a summarized or highly focused portion of the organization's data is placed in a separate database for a specific population of users.
- For example, a company might develop marketing and sales data marts to deal with customer information. Bookseller Barnes & Noble used to maintain a series of data marts—one for point-of-sale data in retail stores, another for college bookstore sales, and a third for online sales.

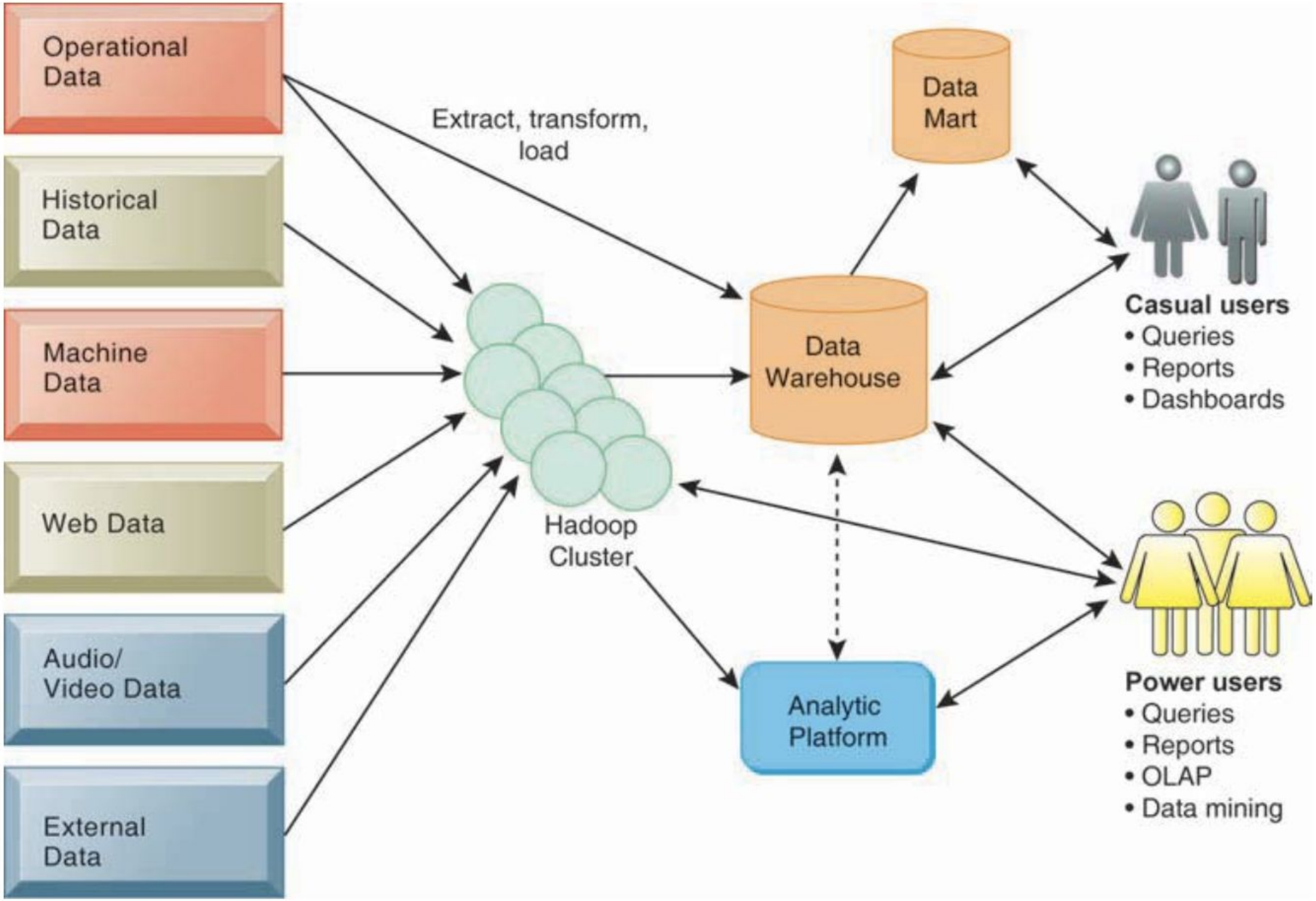
HADOOP

- Relational DBMS and data warehouse products are not well-suited for organizing and analyzing big data or data that do not easily fit into columns and rows used in their data models.
- For handling unstructured and semi-structured data in vast quantities, as well as structured data, organizations are using Hadoop.
- Hadoop is an open source software framework managed by the Apache Software Foundation that enables distributed parallel processing of huge amounts of data across inexpensive computers.
- It breaks a big data problem down into sub-problems, distributes them among up to thousands of inexpensive computer processing nodes, and then combines the result into a smaller data set that is easier to analyze.

IN-MEMORY COMPUTING

- Another way of facilitating big data analysis is to use in-memory computing, which relies primarily on a computer's main memory (RAM) for data storage. (Conventional DBMS use disk storage systems.)
- Users access data stored in system primary memory, thereby eliminating bottlenecks from retrieving and reading data in a traditional, disk-based database and dramatically shortening query response times.
- In-memory processing makes it possible for very large sets of data, amounting to the size of a data mart or small data warehouse, to reside entirely in memory.
- Complex business calculations that used to take hours or days are able to be completed within seconds, and this can even be accomplished on handheld devices.

FIGURE 6.12 COMPONENTS OF A DATA WAREHOUSE



ANALYTICAL TOOLS: RELATIONSHIPS, PATTERNS, TRENDS

- **Online Analytical Processing (OLAP)** supports multidimensional data analysis, enabling users to view the same data in different ways using multiple dimensions.
- **Data mining** provides insights into corporate data that cannot be obtained with OLAP by finding hidden patterns and relationships in large databases and inferring rules from them to predict future behavior.
- **Text mining** tools are now available to help businesses analyze these data. These tools are able to extract key elements from unstructured big data sets, discover patterns and relationships, and summarize the information.
- **Sentiment analysis** software is able to mine text comments in an e-mail message, blog, social media conversation, or survey form to detect favorable and unfavorable opinions about specific subjects.
- The discovery and analysis of useful patterns and information from the World Wide Web is called **Web mining.**

MANAGING DATA RESOURCES

- ESTABLISHING AN INFORMATION POLICY
- An information policy specifies the organization's rules for sharing, disseminating, acquiring, standardizing, classifying, and inventorying information. Information policy lays out specific procedures and account- abilities, identifying which users and organizational units can share information, where information can be distributed, and who is responsible for updating and maintaining the information
- For example, a typical information policy would specify that only selected members of the payroll and human resources department would have the right to change and view sensitive employee data, such as an employee's salary or social security number, and that these departments are responsible for making sure that such employee data are accurate.

- **Data administration** is responsible for the specific policies and procedures through which data can be managed as an organizational resource. These responsibilities include developing information policy, planning for data, overseeing logical database design and data dictionary development, and monitoring how information systems specialists and end-user groups use data.
- The term **data governance** used to describe many of these activities. Promoted by IBM, data governance deals with the policies and processes for managing the availability, usability, integrity, and security of the data employed in an enterprise, with special emphasis on promoting privacy, security, data quality, and compliance with government regulations.
- A large organization will also have a database design and management group within the corporate information systems division that is responsible for defining and organizing the structure and content of the database, and maintaining the database. In close cooperation with users, the design group establishes the physical database, the logical relations among elements, and the access rules and security procedures. The functions it performs are called **database administration**.

ENSURING DATA QUALITY

- Analysis of data quality often begins with a **data quality audit**, which is a structured survey of the accuracy and level of completeness of the data in an information system. Data quality audits can be performed by surveying entire data files, surveying samples from data files, or surveying end users for their perceptions of data quality.
- **Data cleansing**, also known as **data scrubbing**, consists of activities for detecting and correcting data in a database that are incorrect, incomplete, improperly formatted, or redundant. Data cleansing not only corrects errors but also enforces consistency among different sets of data that originated in separate information systems. Specialized data-cleansing software is available to automatically survey data files, correct errors in the data, and integrate the data in a consistent company-wide format.

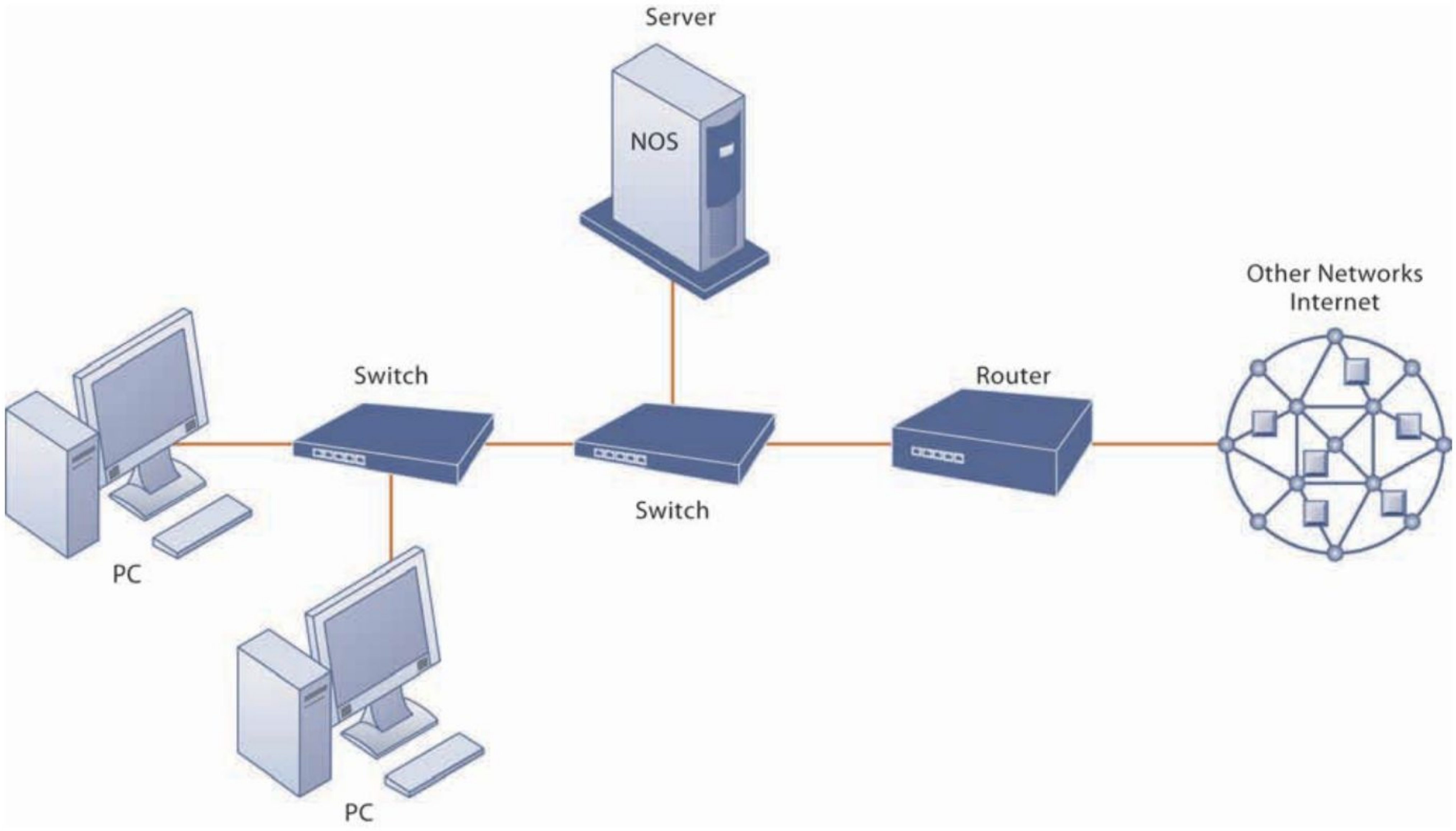
TELECOMMUNICATION

- NETWORKING AND COMMUNICATION TRENDS
- Telecommunications providers today, offer data transmission, Internet access, cellular telephone service, and television programming as well as voice service.
- Cable companies, such as Cablevision and Comcast, offer voice service and Internet access. Computer networks have expanded to include Internet telephone and video services.
- Increasingly, all of these voice, video, and data communications are based on Internet technology.
- Increasingly, voice and data communication, as well as Internet access, are taking place over broadband wireless platforms, such as cell phones, mobile handheld devices, and PCs in wireless networks

WHAT IS A COMPUTER NETWORK?

- If you had to connect the computers for two or more employees together in the same office, you would need a computer network. Exactly what is a network? In its simplest form, a network consists of two or more connected computers.
- Figure 7.1 illustrates the major hardware, software, and transmission components used in a simple network: a client computer and a dedicated server computer, network interfaces, a connection medium, network operating system software, and either a hub or a switch.
- Each computer on the network contains a network interface device to link the computer to the network. The connection medium for linking network components can be a telephone wire, coaxial cable, or radio signal in the case of cell phone and wireless local area networks (Wi-Fi networks).

FIGURE 7.1 COMPONENTS OF A SIMPLE COMPUTER NETWORK



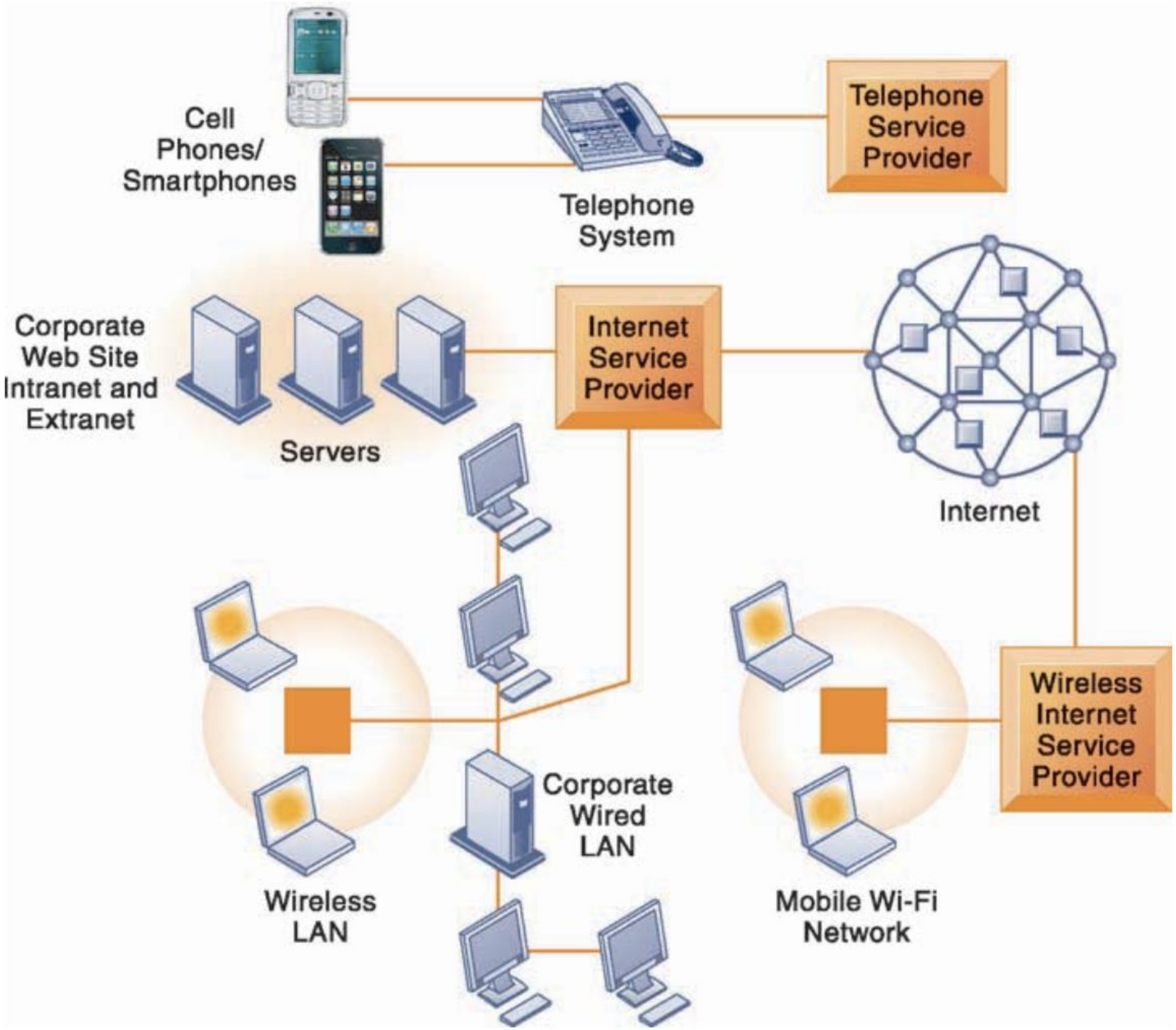
Illustrated here is a very simple computer network, consisting of computers, a network operating system (NOS) residing on a dedicated server computer, cable (wiring) connecting the devices, switches, and a router.

COMPONENTS OF COMPUTER NETWORK

- The **network operating system (NOS)** routes and manages communications on the network and coordinates network resources. It can reside on every computer in the network, or it can reside primarily on a dedicated server computer for all the applications on the network.
- A **server** computer is a computer on a network that performs important network functions for client computers, such as serving up Web pages, storing data, and storing the network operating system (and hence controlling the network).
- Server software such as Microsoft Windows Server, Linux, and Novell Open Enterprise Server are the most widely used network operating systems.

- **Hubs** are very simple devices that connect network components, sending a packet of data to all other connected devices.
- A **switch** has more intelligence than a hub and can filter and forward data to a specified destination on the network.
- A **router** is a communications processor used to route packets of data through different networks, ensuring that the data sent gets to the correct address.
- Network switches and routers have proprietary software built into their hardware for directing the movement of data on the network. This can create network bottlenecks and makes the process of configuring a network more complicated and time-consuming. **Software-defined networking (SDN)** is a new networking approach in which many of these control functions are man- aged by one central program, which can run on inexpensive commodity servers that are separate from the network devices themselves. This is especially helpful in a cloud computing environment with many different pieces of hardware because it allows a network administrator to manage traffic loads in a flexible and more efficient manner.

FIGURE 7.2 CORPORATE NETWORK INFRASTRUCTURE



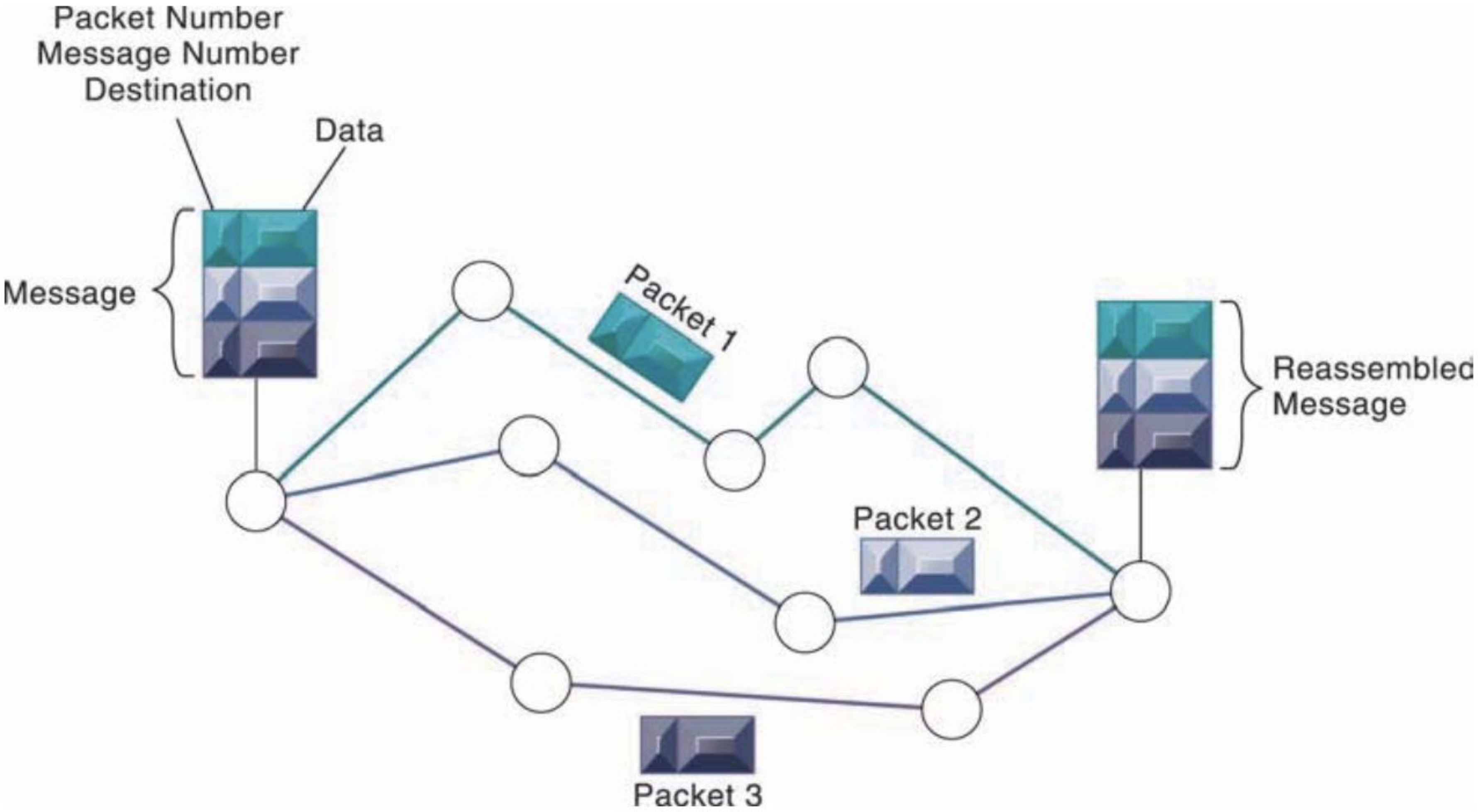
KEY DIGITAL NETWORKING TECHNOLOGIES

- Client/Server Computing
- Packet Switching
- TCP/IP and Connectivity

Transmission Control Protocol/Internet Protocol

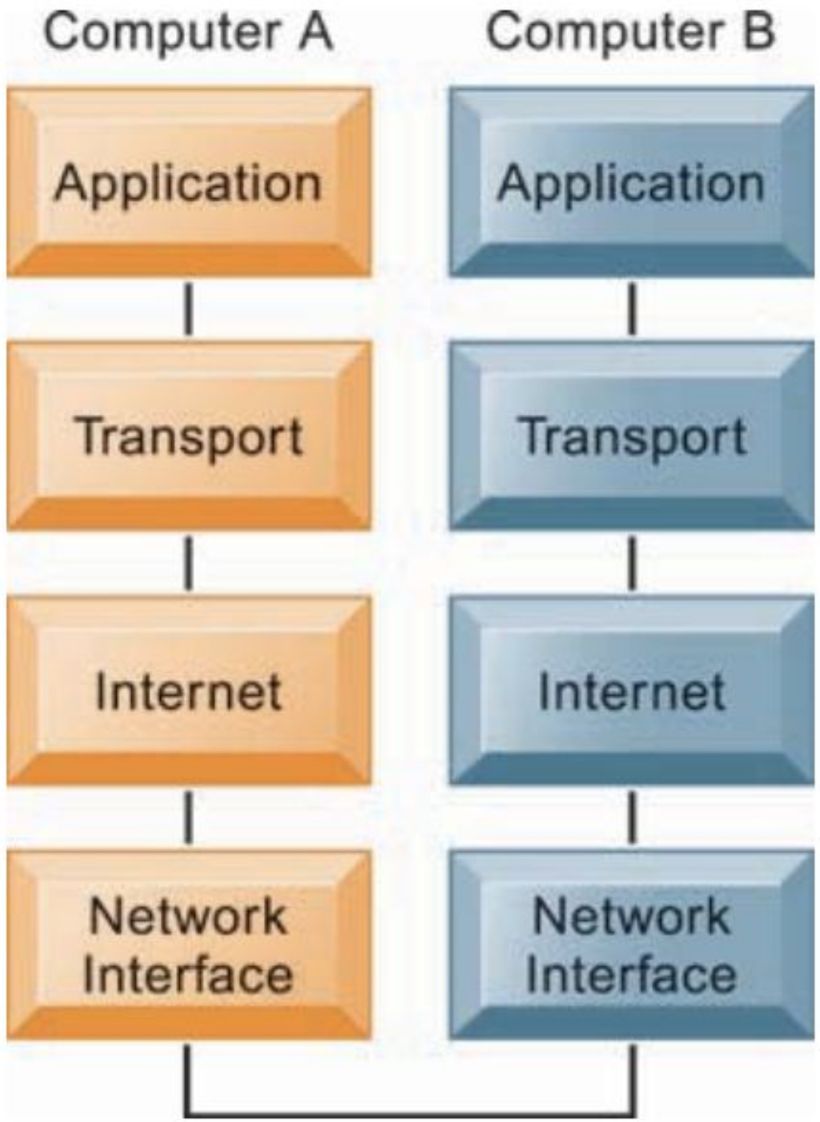
A protocol is a set of rules and procedures governing transmission of information between two points in a network.

FIGURE 7.3 PACKED-SWITCHED NETWORKS AND PACKET COMMUNICATIONS



Data are grouped into small packets, which are transmitted independently over various communications channels and reassembled at their final destination.

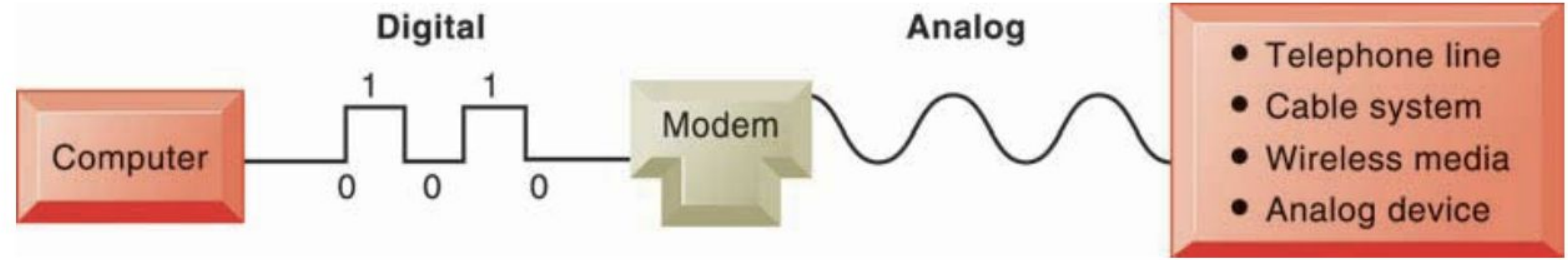
FIGURE 7.4 THE TRANSMISSION CONTROL PROTOCOL/INTERNET PROTOCOL (TCP/IP) REFERENCE MODEL



SIGNALS: DIGITAL VS. ANALOG

- There are two ways to communicate a message in a network: either using an analog signal or a digital signal.
- An analog signal is represented by a continuous waveform that passes through a communications medium and has been used for voice communication.
- The most common analog devices are the telephone handset, the speaker on your computer, or your iPod earphone, all of which create analog waveforms that your ear can hear.
- A digital signal is a discrete, binary waveform, rather than a continuous waveform.
- Digital signals communicate information as strings of two discrete states: one bit and zero bits, which are represented as on-off electrical pulses.
- Computers use digital signals and require a modem to convert these digital signals into analog signals that can be sent over (or received from) telephone lines, cable lines, or wireless media that use analog signals (see Figure 7.5). Modem stands for modulator-demodulator.

FIGURE 7.5 FUNCTIONS OF THE MODEM



A modem is a device that translates digital signals into analog form (and vice versa) so that computers can transmit data over analog networks such as telephone and cable networks.

TABLE 7.1 TYPES OF NETWORKS

TYPE	AREA
Local area network (LAN)	Up to 500 meters (half a mile); an office or floor of a building
Campus area network (CAN)	Up to 1,000 meters (a mile); a college campus or corporate facility
Metropolitan area network (MAN)	A city or metropolitan area
Wide area network (WAN)	A transcontinental or global area

TABLE 7.2 PHYSICAL TRANSMISSION MEDIA

TRANSMISSION MEDIUM	DESCRIPTION	SPEED
Twisted pair wire (CAT 5)	Strands of copper wire twisted in pairs for voice and data communications. CAT 5 is the most common 10 Mbps LAN cable. Maximum recommended run of 100 meters.	10 Mbps to 1 Gbps
Coaxial cable	Thickly insulated copper wire, which is capable of high-speed data transmission and less subject to interference than twisted wire. Currently used for cable TV and for networks with longer runs (more than 100 meters).	Up to 1 Gbps
Fiber optic cable	Strands of clear glass fiber, transmitting data as pulses of light generated by lasers. Useful for high-speed transmission of large quantities of data. More expensive than other physical transmission media and harder to install; often used for network backbone.	500 Kbps to 6+Tbps
Wireless transmission media	Based on radio signals of various frequencies and includes both terrestrial and satellite microwave systems and cellular networks. Used for long-distance, wireless communication and Internet access.	Up to 600+ Mbps

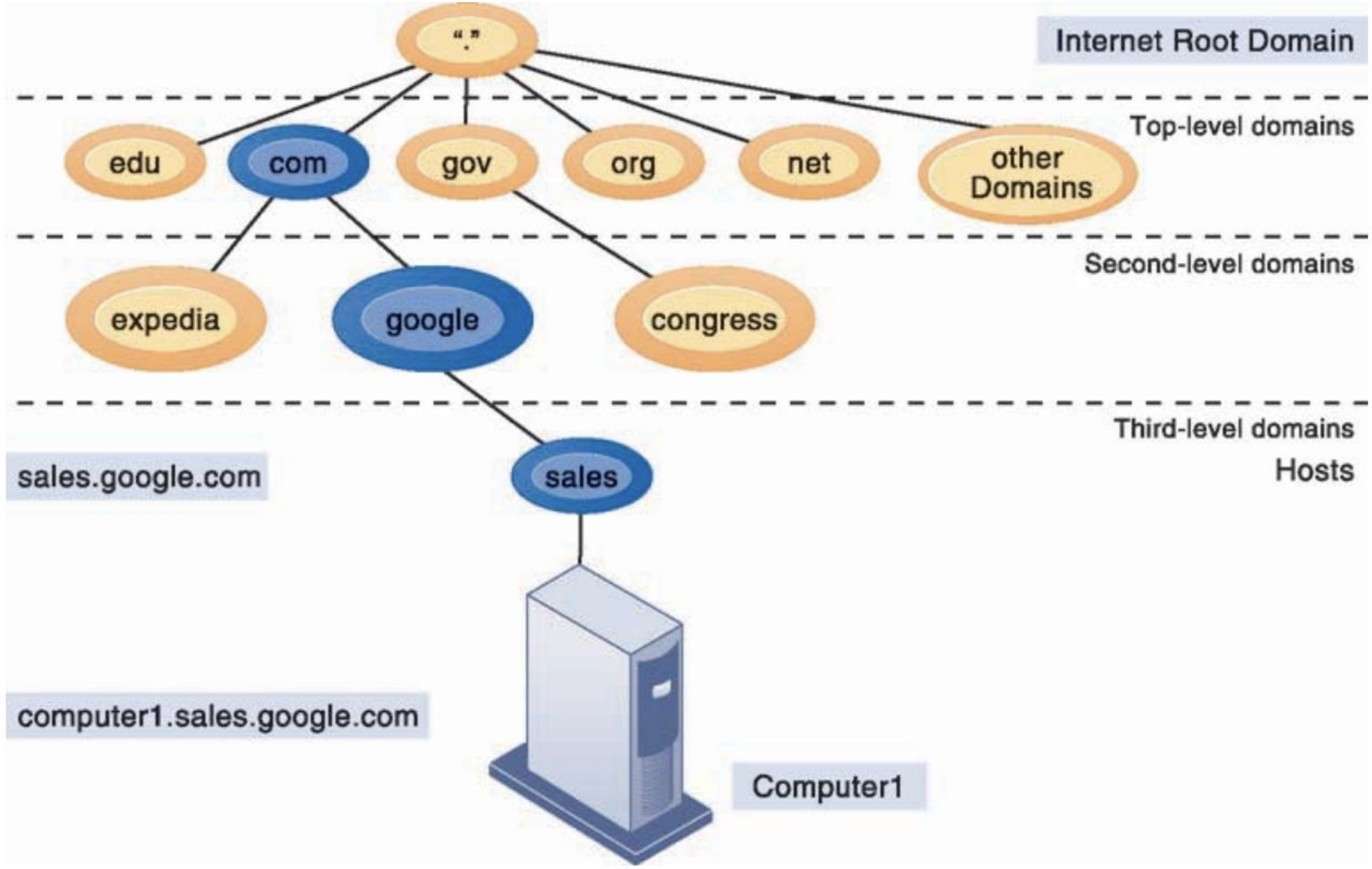
WHAT IS THE INTERNET?

- The Internet is a vast network that connects computers all over the world. Through the Internet, people can share information and communicate from anywhere with an Internet connection.
- Most homes and small businesses connect to the Internet by subscribing to an Internet service provider. An Internet service provider (ISP) is a commercial organization with a permanent connection to the Internet that sells temporary connections to retail subscribers.
- The Internet is based on the TCP/IP networking protocol suite described earlier in this chapter.
- Every computer on the Internet is assigned a unique Internet Protocol (IP) address, which currently is a 32-bit number represented by four strings of numbers ranging from 0 to 255 separated by periods.
- For instance, the IP address of www.microsoft.com is 207.46.250.119.

THE DOMAIN NAME SYSTEM

- Because it would be incredibly difficult for Internet users to remember strings of 12 numbers, the Domain Name System (DNS) converts domain names to IP addresses.
- The domain name is the English-like name that corresponds to the unique 32-bit numeric IP address for each computer connected to the Internet. DNS servers maintain a database containing IP addresses mapped to their corresponding domain names. To access a computer on the Internet, users need only specify its domain name.

FIGURE 7.6 THE DOMAIN NAME SYSTEM



Domain Name System is a hierarchical system with a root domain, top-level domains, second-level domains, and host computers at the third level.

DOMAIN EXTENSIONS

.com	Commercial organizations/businesses
.edu	Educational institutions
.gov	U.S. government agencies
.mil	U.S. military
.net	Network computers
.org	Nonprofit organizations and foundations
.biz	Business firms
.info	Information providers

TABLE 7.3 MAJOR INTERNET SERVICES

CAPABILITY	FUNCTIONS SUPPORTED
E-mail	Person-to-person messaging; document sharing
Chatting and instant messaging	Interactive conversations
Newsgroups	Discussion groups on electronic bulletin boards
Telnet	Logging on to one computer system and doing work on another
File Transfer Protocol (FTP)	Transferring files from computer to computer
World Wide Web	Retrieving, formatting, and displaying information (including text, audio, graphics, and video) using hypertext links

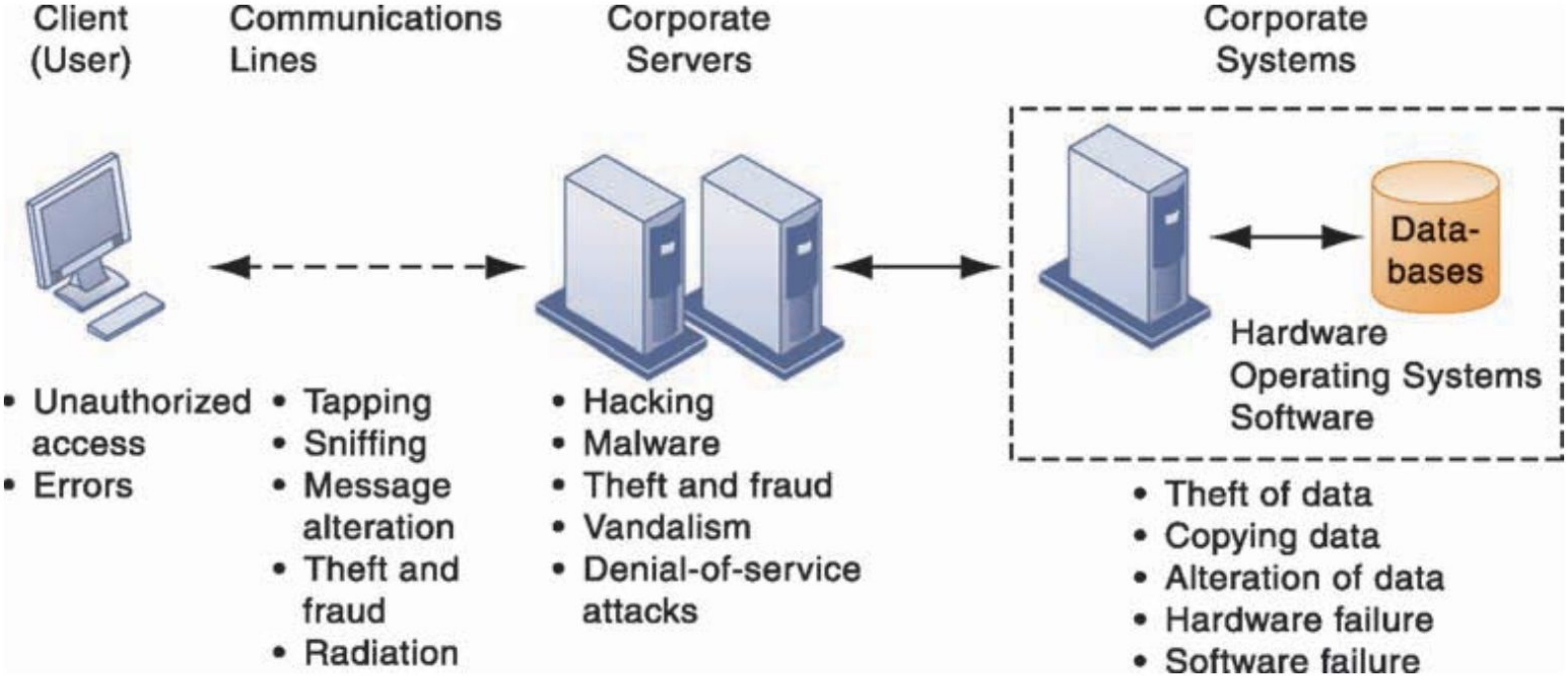
THE WIRELESS REVOLUTION

- CELLULAR SYSTEMS
  - ✔ 3G
  - ✔ 4G
  - ✔ 5G
- WIRELESS COMPUTER NETWORKS AND INTERNET ACCESS
  - ✔ Bluetooth
  - ✔ Wi-Fi
  - ✔ Radio Frequency Identification (RFID)

WHY SYSTEMS ARE VULNERABLE

- When large amounts of data are stored in electronic form, they are vulnerable to many more kinds of threats than when they existed in manual form.
- Through communications networks, information systems in different locations are interconnected.
- The potential for unauthorized access, abuse, or fraud is not limited to a single location but can occur at any access point in the network.

FIGURE 8.1 CONTEMPORARY SECURITY CHALLENGES AND VULNERABILITIES



The architecture of a Web-based application typically includes a Web client, a server, and corporate information systems linked to databases. Each of these components presents security challenges and vulnerabilities. Floods, fires, power failures, and other electrical problems can cause disruptions at any point in the network.

INTERNET VULNERABILITIES

- Large public networks, such as the Internet, are more vulnerable than internal networks because they are virtually open to anyone.
- The Internet is so huge that when abuses do occur, they can have an enormously widespread impact. When the Internet becomes part of the corporate network, the organization's information systems are even more vulnerable to actions from outsiders.
- Computers that are constantly connected to the Internet by cable modems or digital subscriber line (DSL) lines are more open to penetration by outsiders because they use fixed Internet addresses where they can be easily identified. (With dial-up service, a temporary Internet address is assigned for each session.)
- A fixed Internet address creates a fixed target for hackers.

WIRELESS SECURITY CHALLENGES

- Is it safe to log onto a wireless network at an airport, library, or other public location?
- Even the wireless network in your home is vulnerable because radio frequency bands are easy to scan.
- Both Bluetooth and Wi-Fi networks are susceptible to hacking by eavesdroppers.
- Local area networks (LANs) using the 802.11 standard can be easily penetrated by out- siders armed with laptops, wireless cards, external antennae, and hacking software.
- Hackers use these tools to detect unprotected networks, monitor network traffic, and, in some cases, gain access to the Internet or to corporate networks.

MALICIOUS SOFTWARE:VIRUSES,WORMS,TROJAN HORSES, AND SPYWARE

- Malicious software programs are referred to as malware and include a variety of threats, such as computer viruses, worms, and Trojan horses.
- A computer **virus** is a rogue software program that attaches itself to other software programs or data files in order to be executed, usually without user knowledge or permission. Most computer viruses deliver a “payload.”
- The payload such as instructions to display a message or image, or it may be highly destructive—destroying programs or data, clogging computer memory, reformatting a computer’s hard drive, or causing programs to run improperly.
- Viruses typically spread from computer to computer when humans take an action, such as sending an e-mail attachment or copying an infected file.

WORMS

- Most recent attacks have come from **worms**, which are independent computer programs that copy themselves from one computer to other computers over a network.
- Unlike viruses, worms can operate on their own without attaching to other computer program files and rely less on human behavior in order to spread from computer to computer.
- This explains why computer worms spread much more rapidly than computer viruses. Worms destroy data and programs as well as disrupt or even halt the operation of computer networks.

TROJAN HORSES

- A **Trojan horse** is a software program that appears to be normal but then does something other than expected.
- The Trojan horse is not itself a virus because it does not replicate, but it is often a way for viruses or other malicious code to be introduced into a computer system.
- The term Trojan horse is based on the huge wooden horse used by the Greeks to trick the Trojans into opening the gates to their fortified city during the Trojan War. Once inside the city walls, Greek soldiers hidden in the horse revealed themselves and captured the city.

SPYWARE

- **Spyware** is any software that installs itself on your computer and starts covertly monitoring your online behavior without your knowledge or permission. Spyware is a kind of malware that secretly gathers information about a person or organization and relays this data to other parties.

HACKERS AND COMPUTER CRIME

- A hacker is an individual who intends to gain unauthorized access to a computer system
- Hacker activities have broadened beyond mere system intrusion to include theft of goods and information, as well as system damage and cybervandalism, the intentional disruption, defacement, or even destruction of a Web site or corporate information system.

SPOOFING AND SNIFFING

- Spoofing also may involve redirecting a Web link to an address different from the intended one, with the site masquerading as the intended destination. For example, if hackers redirect customers to a fake Web site that looks almost exactly like the true site, they can then collect and process orders, effectively stealing business as well as sensitive customer information from the true site.
- A sniffer is a type of eavesdropping program that monitors information traveling over a network. When used legitimately, sniffers help identify potential network trouble spots or criminal activity on networks, but when used for criminal purposes, they can be damaging and very difficult to detect. Sniffers enable hackers to steal proprietary information from anywhere on a network, including e-mail messages, company files, and confidential reports.

BUSINESS VALUE OF SECURITY AND CONTROL

- Lack of sound security and control can cause firms relying on computer systems for their core business functions to lose sales and productivity. Information assets, such as confidential employee records, trade secrets, or business plans, lose much of their value if they are revealed to outsiders or if they expose the firm to legal liability.
- New laws, such as HIPAA, the Sarbanes-Oxley Act, and the Gramm-Leach-Bliley Act, require companies to practice stringent electronic records management and adhere to strict standards for security, privacy, and control. Legal actions requiring electronic evidence and computer forensics also require firms to pay more attention to security and electronic records management.

ELECTRONIC EVIDENCE

Digital data stored on

- portable storage devices,
- CDs, computer hard disk drives,
- as well as in e-mail,
- instant messages, and
- e-commerce transactions over the Internet.

E-mail is currently the most common type of electronic evidence.

COMPUTER FORENSICS

- Computer forensics is the scientific collection, examination, authentication, preservation, and analysis of data held on or retrieved from computer storage media in such a way that the information can be used as evidence in a court of law. It deals with the following problems:
  - Recovering data from computers while preserving evidential integrity
  - Securely storing and handling recovered electronic data
  - Finding significant information in a large volume of electronic data
  - Presenting the information to a court of law